Interleaving for slow flat fading channels

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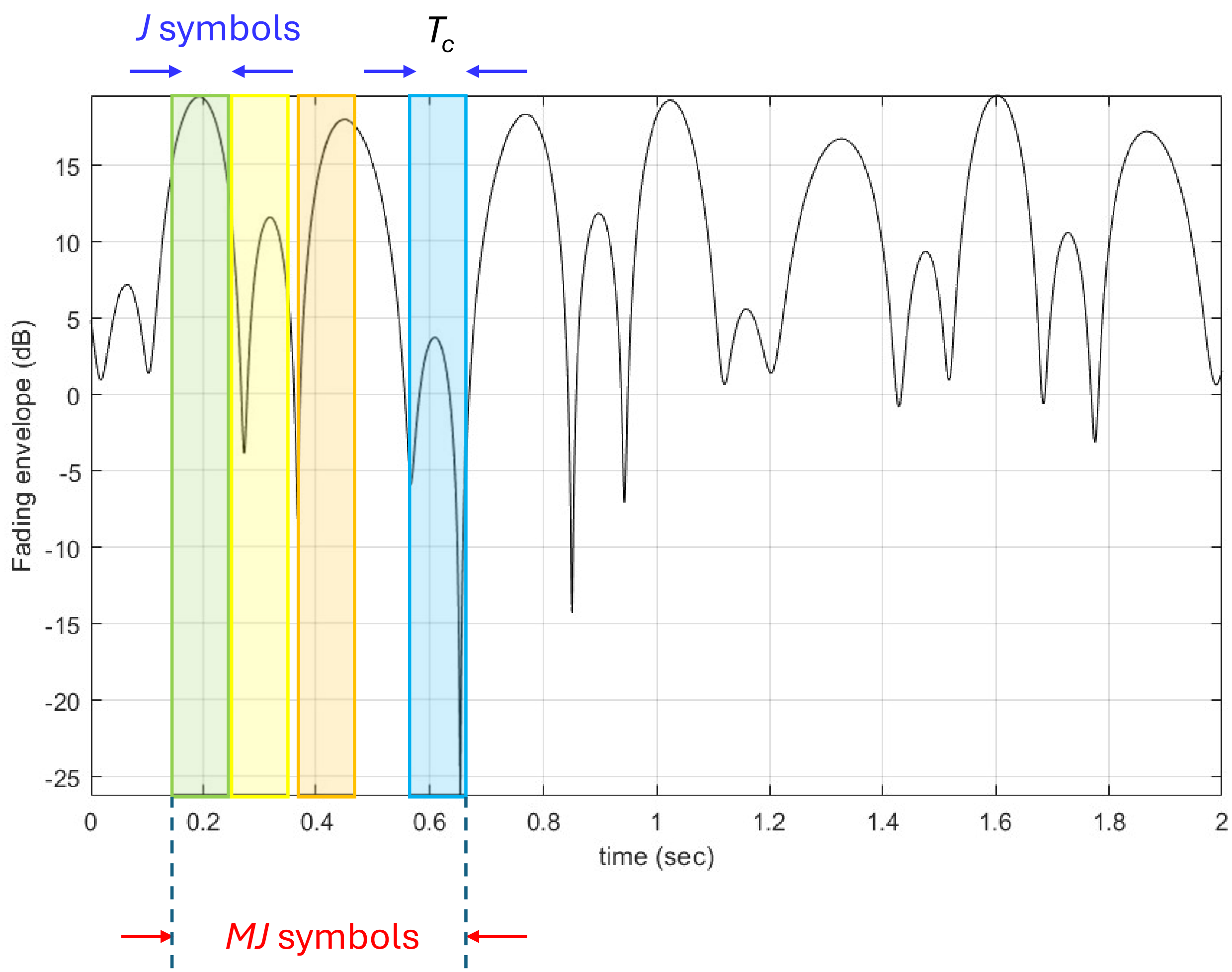
Slow Rayleigh fading (Jakes' model, f_m=5 Hz)

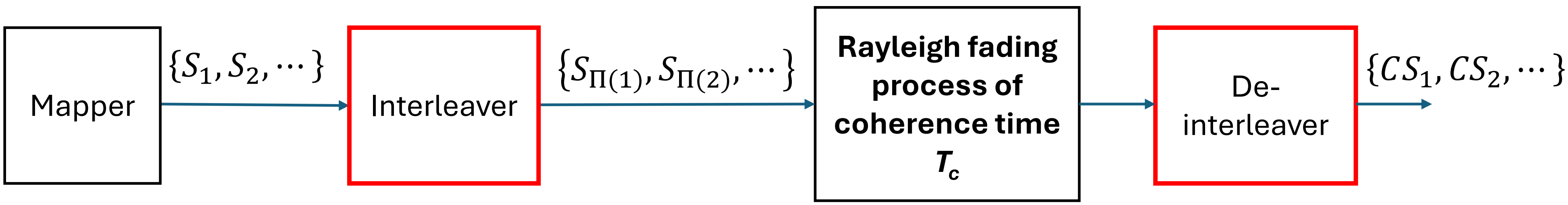


Colors used for different received signal energy levels

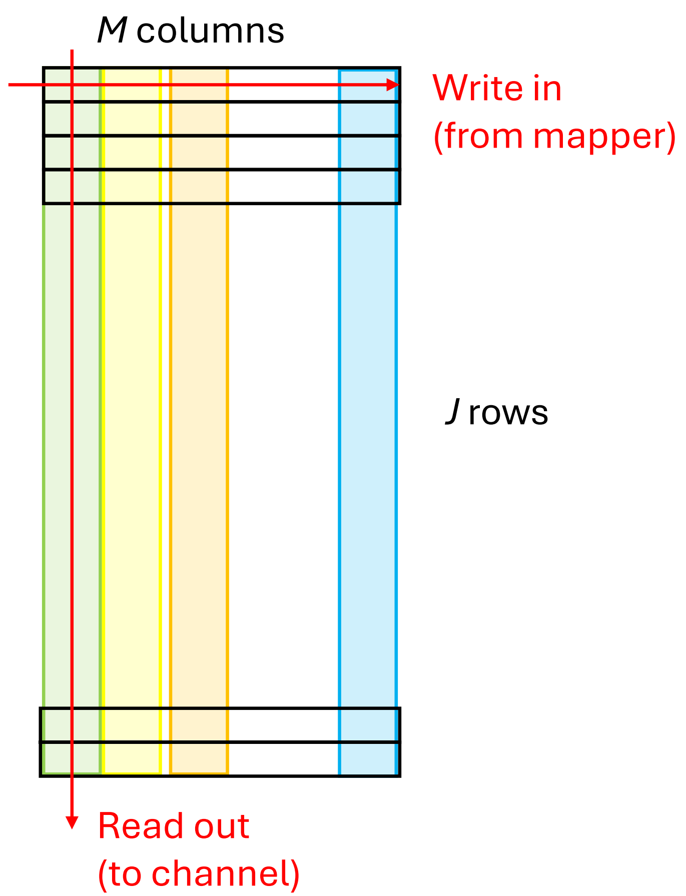
Why interleaving?

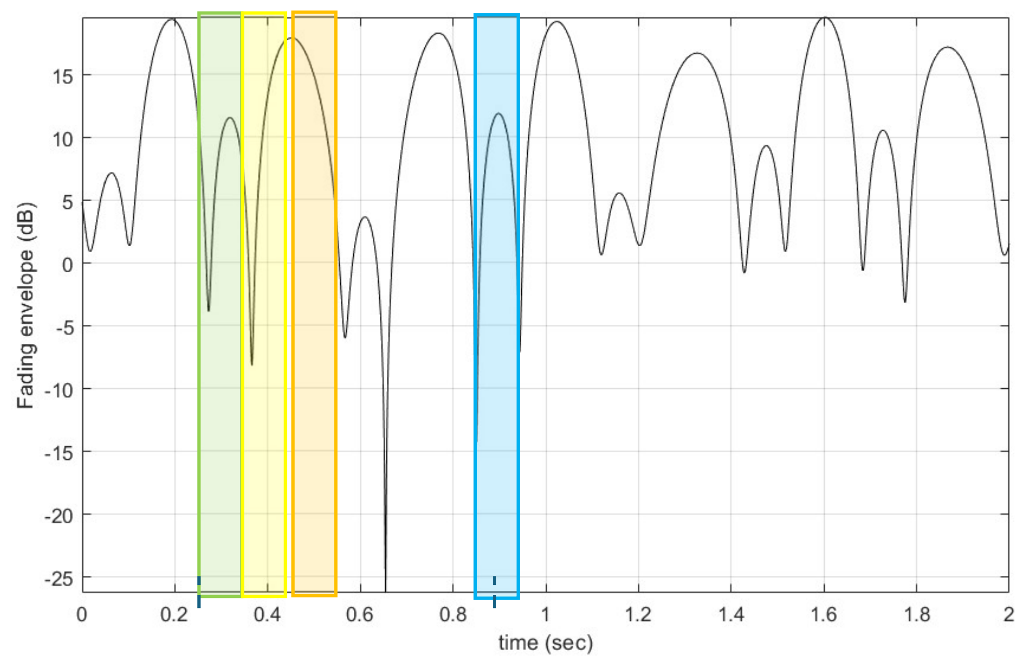
- The goal is to scramble (***permute the positions*** of) consecutive correlated symbols so that the received sequence after descrambling contains statistically independent symbols.
- This is referred to in the literature as “breaking the correlation” or “spreading bursts of errors”
- Since ***correlated low energy symbols are spread over time***, interleaving allows error correcting codes to operate efficiently, reducing the average probability of a bit error.
- There are many types of interleavers. For simplicity of exposition, here we consider ***rectangular interleavers***.

Interleaving and de-interleaving

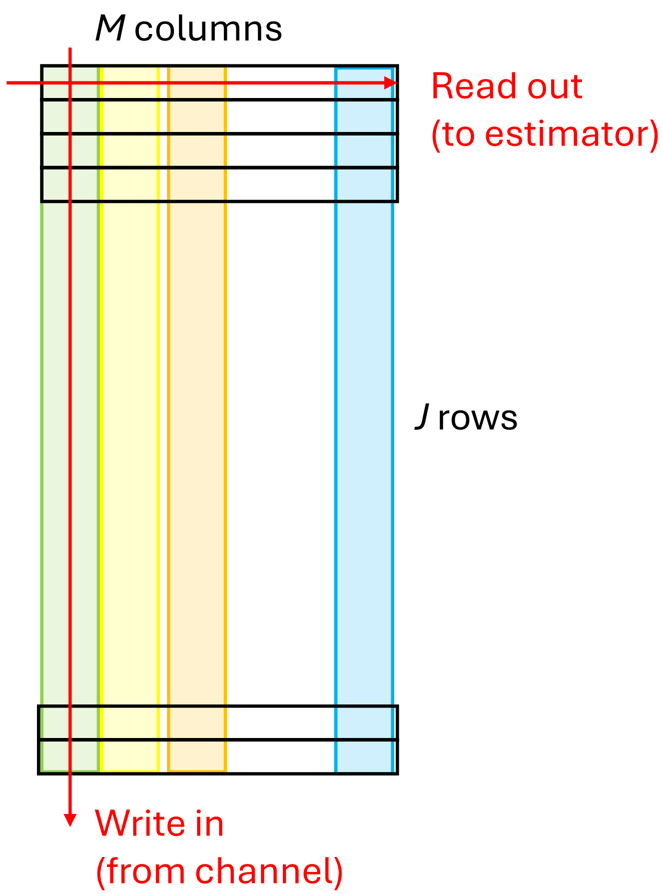


Rectangular interleaver

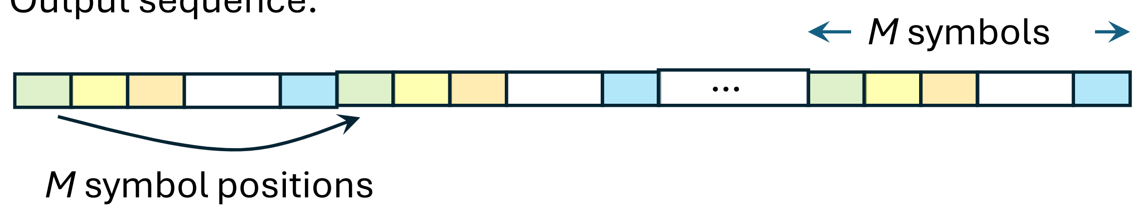




Rectangular de-interleaver



Output sequence:

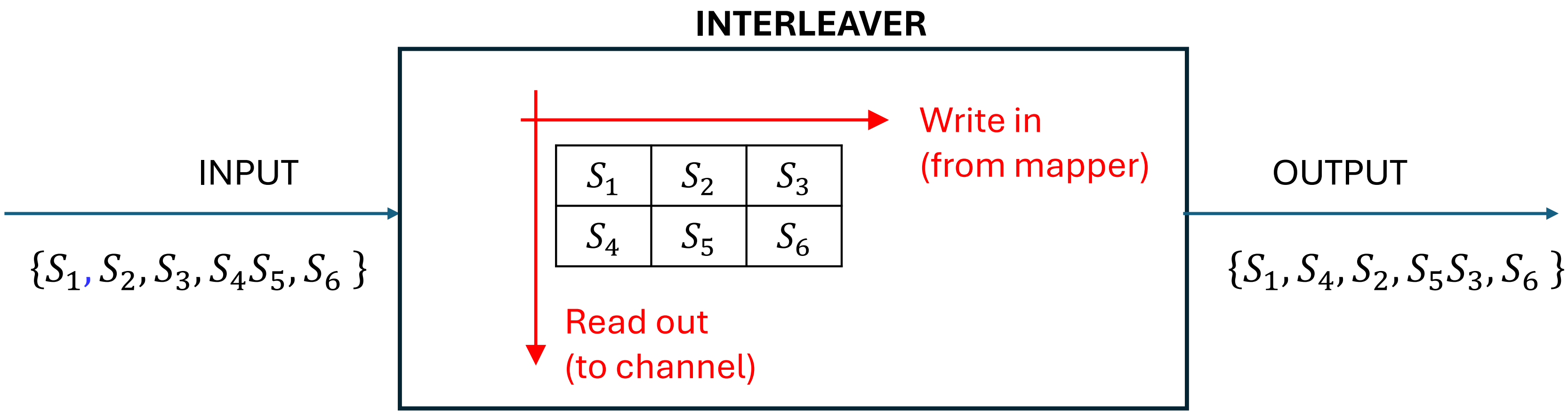


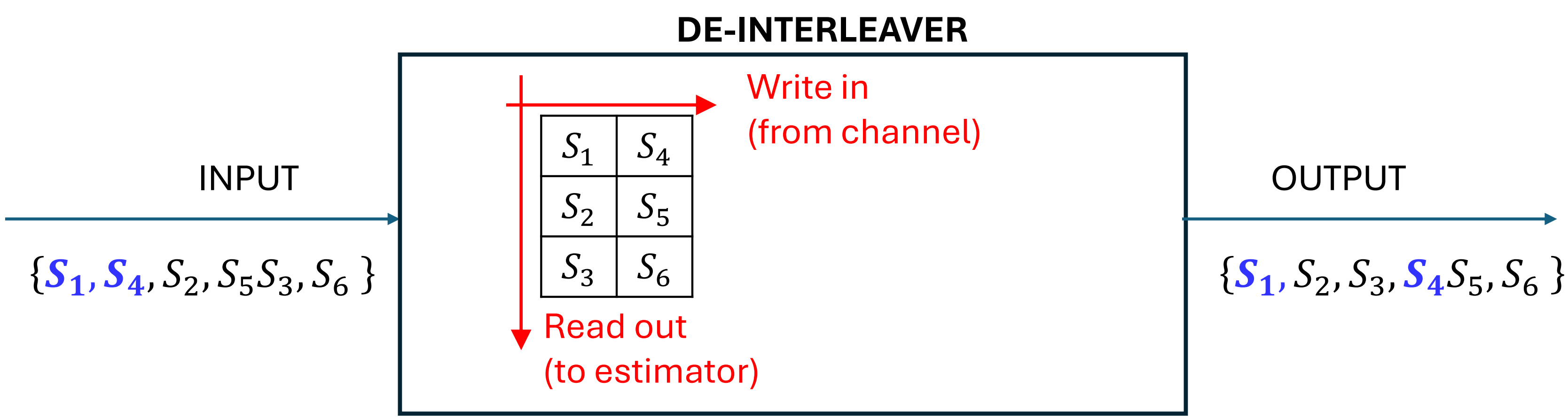
Low-energy symbols are spread over time

(1) This works as long as $JT > T_c$

(2) The choice of interleaver parameter M depends on the error correcting code used….

Example: Rectangular interleaving with J=2, M=3





Example: EPC (3,2,2) code with J=2, M=3